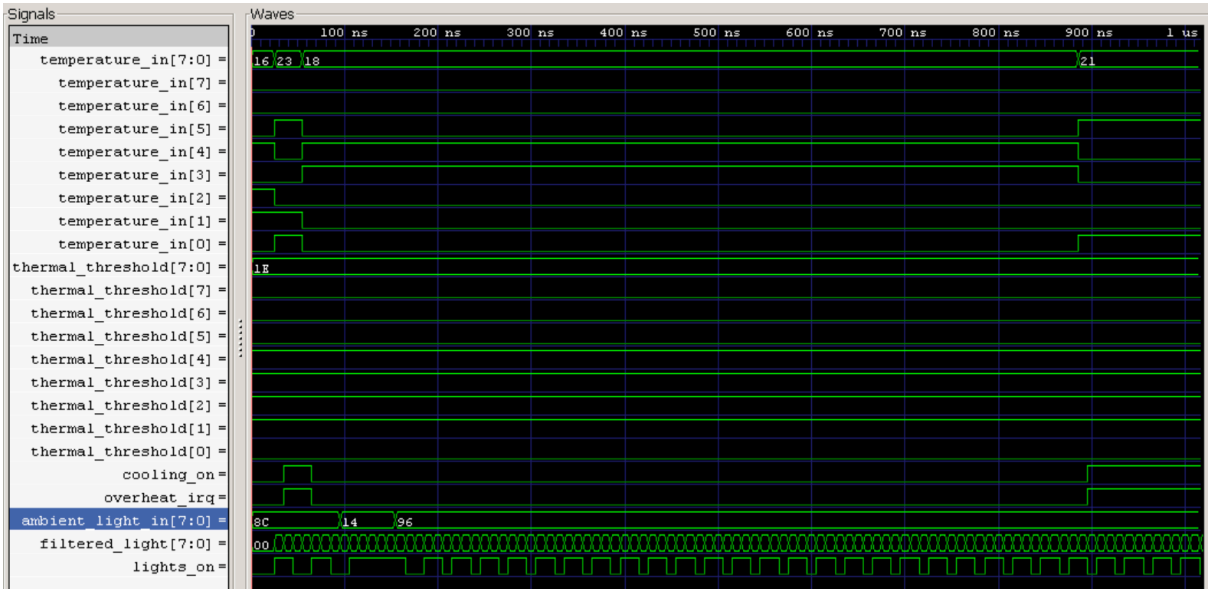
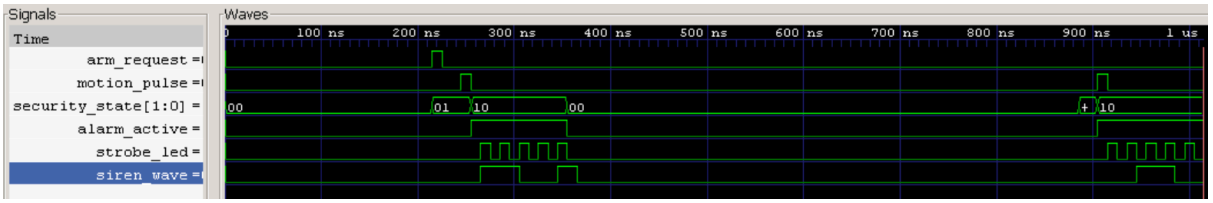
Figure1: Thermal and Lighting Response:



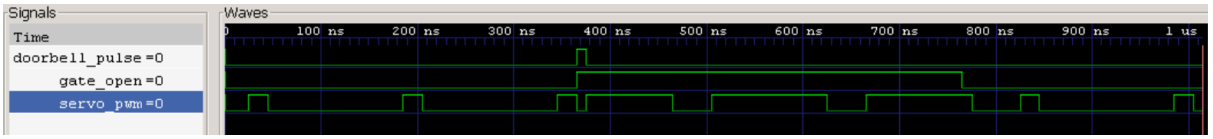
Caption:"When temperature exceeds the threshold, the cooling and overheat interrupt signals assert.When ambient light decreases, the filtered light level falls below threshold and the lighting output turns on**."**

Figure2: Security FSM Operation**:**



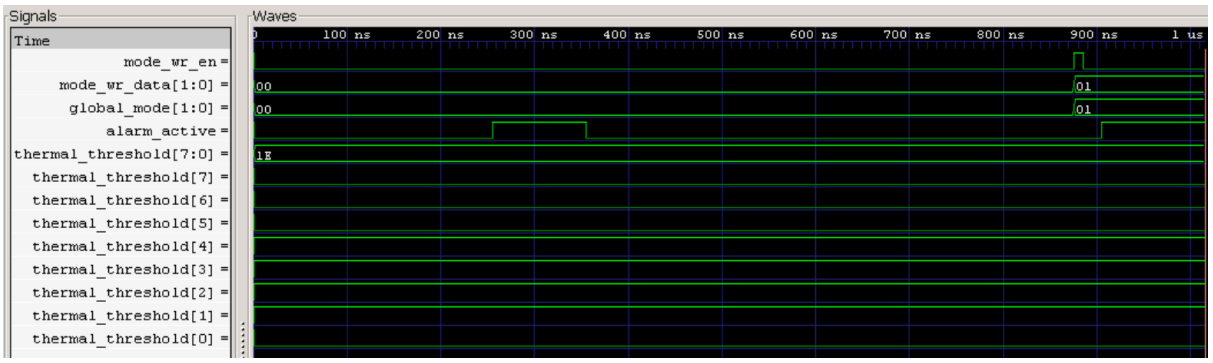
Caption:" The security controller transitions from **NORMAL** to **ARMED** and then to **ALARM** on motion detection. During **ALARM** state the strobe and siren outputs are activated**."**

Figure3: Access control and gate actuation



Caption:"A doorbell pulse opens thee gate and enables **PWM-**based servo control for access actuation**."**

Figure4: Global mode control



Caption:”The Global governance register updates the system Mode from comfort mode to lockdown mode, changing the smart home SoC response behaviour.**”**